

Regularization

- Original loss function by weight decay. ↑ from tree

$$L(a, b) = \frac{1}{N} \sum_{i=1}^N (\hat{y}_i - y_i)^2 = \frac{1}{N} \sum_{i=1}^N \frac{1}{2} (w^T x^{(i)} + b - y^{(i)})^2$$

$$\hat{y}_i = w^T x_i$$

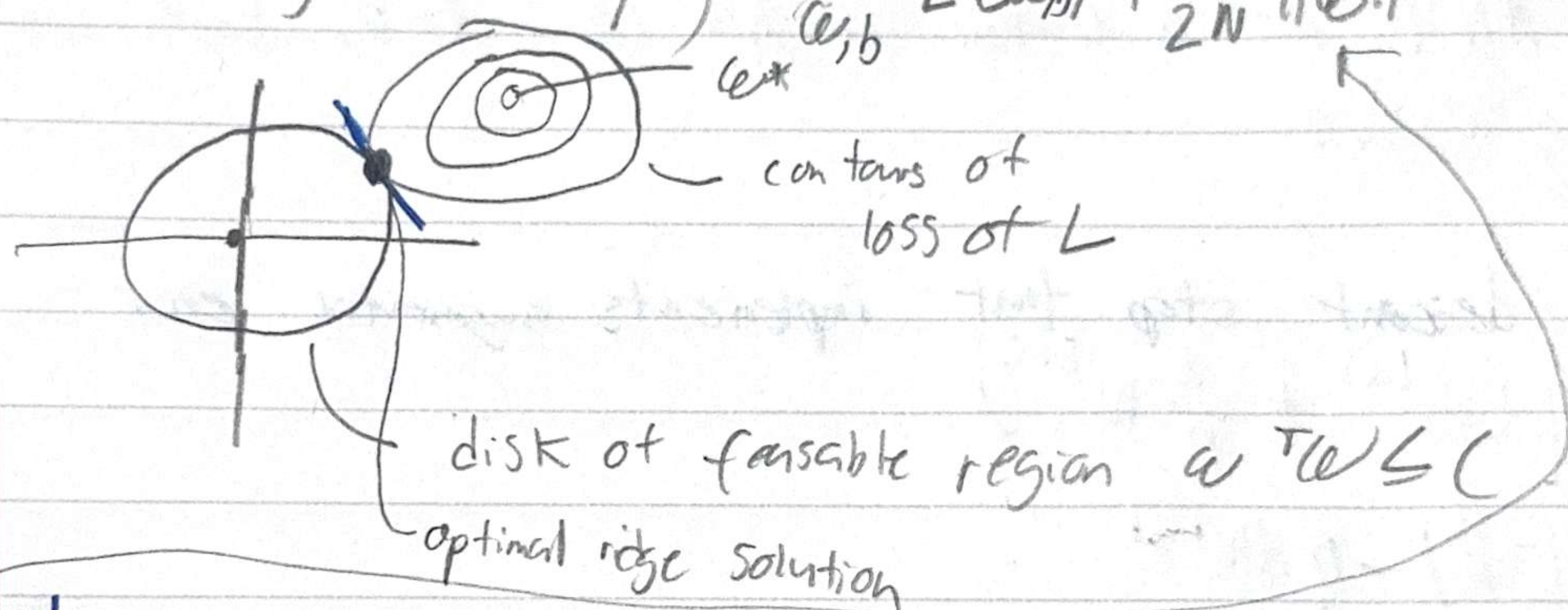
$$* \|a\|_2^2 = \|a\|^2$$

$$a^T a = \|a\|^2$$

Ridge Regression (L2)

- Min L s.t. $w^T w \leq C$

- Same thing as saying min $L(a, b) + \frac{\lambda}{2N} \|w\|^2$



Math

Mathematical derivation that turns Constrained Ridge problem into unconstrained objective that actually must be coded.

$$\frac{dL}{dw}(dw) \quad dw \propto \Delta w \quad \begin{matrix} \nearrow \text{typically } \Delta w = -\frac{dL}{dw} \\ \text{optimization} \end{matrix}, \text{ but we are now in constrained}$$

$$dw = -2\lambda_c w$$

$$dw + 2\lambda_c w = 0 \quad \rightarrow \text{optimal}$$

- at optimal point, gradient of the loss plus gradient of constraint must cancel exactly
- this is the tangent intersection above in blue

$$\text{grad}(L + \lambda_c w^T w) = 0$$

Critical point and solution

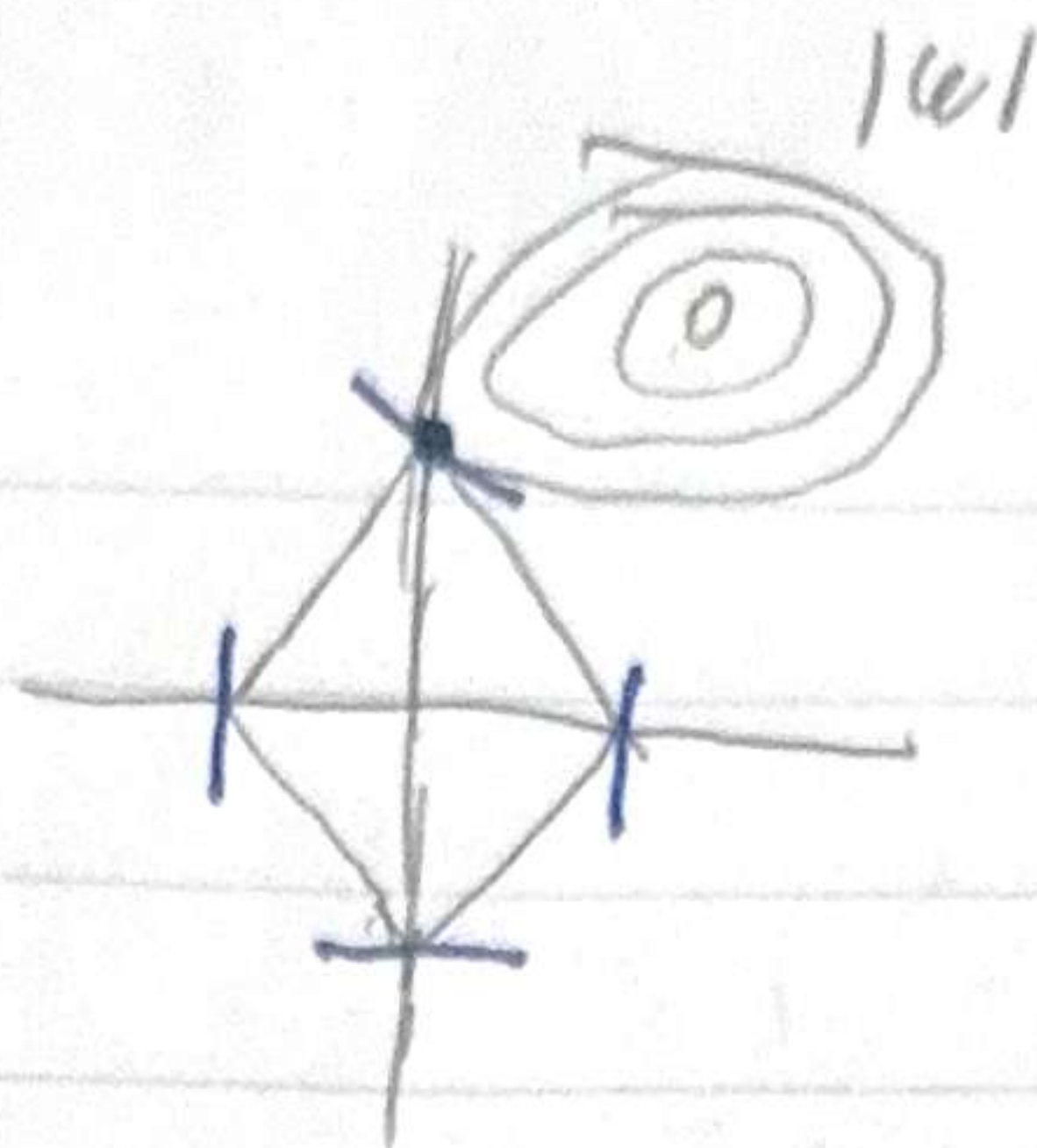
$$\text{of function: } \min_w (L(w) + \lambda_c w^T w)$$

↑ Augmented error function with regularization factored in

Lasso (L1)

~~Regression~~ Regression

$$\sum_{i=1}^n |w_i| \leq t$$



- Same loss ellipse can now touch at a corner for optimal
- When this occurs, one weight corresponding to a feature becomes exactly zero.
removing a feature's effect
- Thus, Lasso does automatic feature selection/sparsity vs ridge shrinking all weights
- Lasso CAN set a feature weight $w_i = 0$
- Ridge CANNOT set a feature weight (w_i) to 0, but can significantly shrink it; $w_i = 0.0001$

New Gradient descent step that implements augmented error

$$\begin{aligned} W^{(L)} &= W^{(L)} - \eta \left(dW^{(L)} + \frac{\lambda}{N} W^{(L)} \right) \\ &= W^{(L)} \left(1 - \frac{\lambda}{N} \right) - \eta dW^{(L)} \end{aligned}$$

Heuristics

We want good accuracy with Great runtime vs. Perfect accuracy with awful runtime

• Typical constraint:

max Accuracy

s.t. Runtime $\leq 10^6$

→ memory